Do Now · retrieval

Von Neumann and Harvard architectures

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is the ideal completion cycle for four independent instructions in a three-stage pipeline?

Q2

What does MDR contain during an instruction fetch?

Q3

What happens in the Decode stage?

Q4

Calculate Sum and Carry-out for a full adder with A=1, B=0, Carry-in=1. Explain in binary.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Von Neumann organisation — Instructions and data share the same address space and a pathway; access can compete for bandwidth.

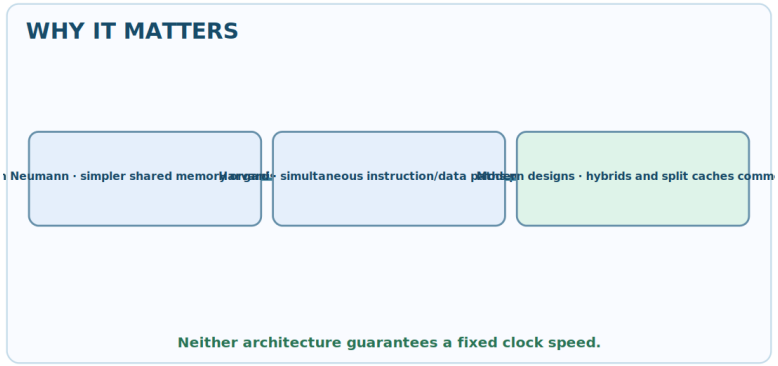
Core idea 2

Harvard organisation — Instruction and data memories/address spaces are separate, allowing independent access paths.

Core idea 3

Recognise contemporary designs — Many processors have aspects of both, such as a unified main memory and separate instruction/data caches (modified Harvard).

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

Von Neumann — Architecture with shared memory space for instructions and data.

Harvard — Architecture with separate program and data storage paths.

bottleneck — Limit caused by a shared resource or transfer path.

instruction cache — Fast cache storing instructions.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

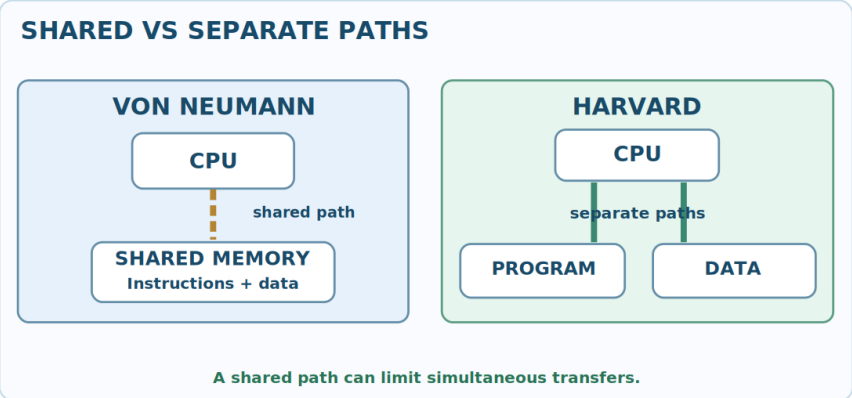
A CPU needs to fetch an instruction while reading its operand. Explain possible contention in a Von Neumann design.

Follow the worked method

1. Predict first — The instruction fetch and data read may both need the same memory transfer pathway.

2. Explain and check — The requests may have to be scheduled rather than served simultaneously; this is related to the Von Neumann bottleneck.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first

Separate program and data memories indicate Harvard-style organisation.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

State where instructions and data are stored in Von Neumann and Harvard architectures.

Question 2

A CPU frequently fetches an instruction while reading data. Explain a potential Von Neumann limitation and Harvard advantage.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

A modern CPU uses unified main memory but separate L1 instruction and data caches. Explain why it cannot be described as a purely classical Harvard or purely classical Von Neumann design.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Every modern processor is exclusively a pure Von Neumann or pure Harvard machine.

Use key vocabulary from memory

1. Explain Von Neumann in your own words.

2. Explain Harvard in your own words.

3. Explain bottleneck in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.